

FERMI-LEVEL FOR INTRINSIC SEMICONDUCTOR.

$$n = N \exp\left(\frac{-\Delta E}{KT}\right)$$

Using the concentration of electrons:

$$n_e = N_c \exp\left(\frac{-(\Delta E_e)}{KT}\right)$$



$$n_e = N_c \exp\left(\frac{-(E_C - E_F)}{KT}\right)$$

Using the concentration of holes:

$$n_e = N_c \exp\left(\frac{-(\Delta E_h)}{KT}\right)$$



$$n_h = N_V \exp\left(\frac{-(E_F - E_V)}{KT}\right)$$



$$n_e = n_h$$



$$N_C \cong N_V$$

$$\exp\left(\frac{-E_C + E_F}{KT}\right) = \exp\left(\frac{-E_F + E_V}{KT}\right)$$



$$\left(\frac{-E_C + E_F}{KT}\right) = \left(\frac{-E_F + E_V}{KT}\right)$$



$$-E_C + E_F = -E_F + E_V$$



$$2E_F = E_C + E_V$$



$$E_F = \frac{E_C + E_V}{2}$$

